<u>Tittle: RESTAURANT MANAGEMENT SYSTEM</u>

Course Code : CSE282.3

Course Title : Introduction to Programming Language II Lab

Department : CSE

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Workplace : SOUTHEAST UNIVERSITY

Assignment Submission Date : 04 June 2026

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<u>Github Link:</u>

https://github.com/Ro453/Restaurent-Mangement-System

Title: RESTAURANT MANAGEMENT SYSTEM

INTRODUCTION

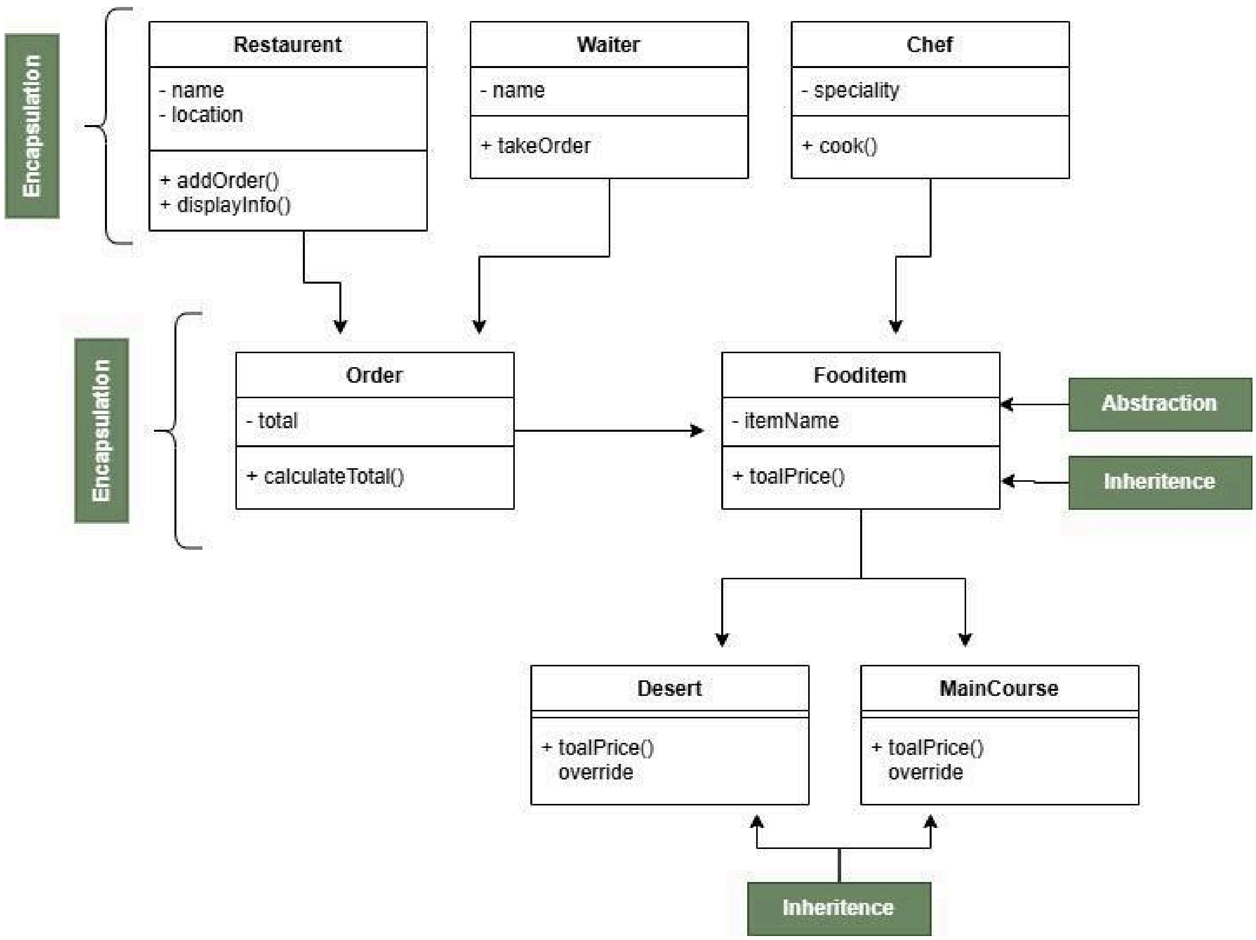
1.Project Objective:

Develop a Java desktop application to simulate daily restaurant operations, allowing staff to add menu items, build customer orders, and track total sales—while demonstrating core OOP principles via a Swing-based GUI.

2.Project Overview:

The system lets a cashier/waiter add food items (Main Course or Dessert) with prices and optional service charges, build orders interactively, submit them for handling by a waiter and chef, and record total sales. It uses a user-friendly Swing interface and models real-world entities like Restaurant, Order, FoodItem, Waiter, and Chef

3. Class Diagram:



4.System Theory (OPP Concepts):

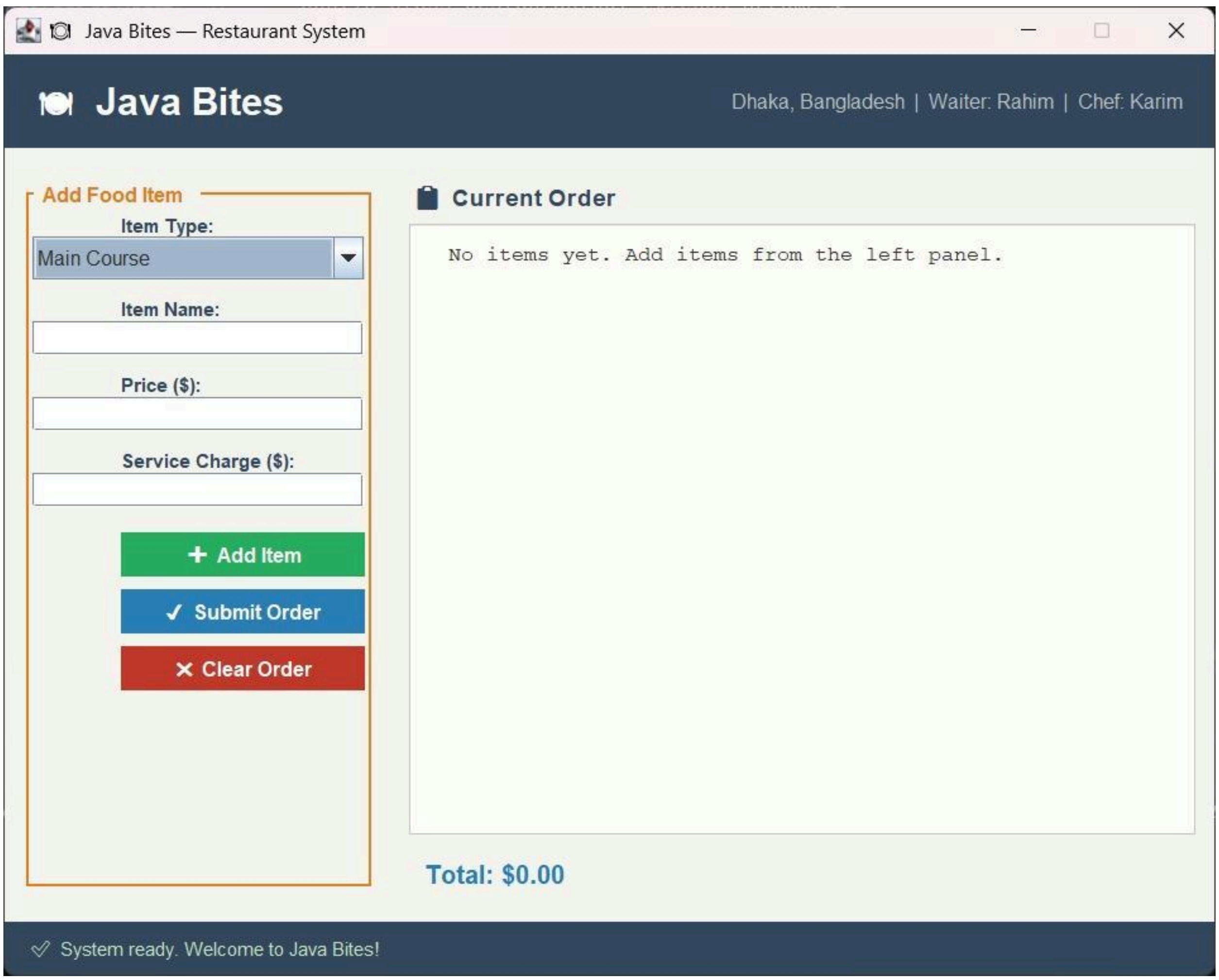
PRINCIPLE	IMPLEMENTATION
Encapsulation	Private fields (e.g., totalSales, orderId, itemPrice) with public getters/controlled methods.
Abstraction	Fooditem abstract class with abstract method totalPrice() – implementation hidden in subclasses.
Inheritance	MainCourse and Dessert extend Fooditem, reusing fields and getters.
Polymorphism	Both subclasses override totalPrice(). Same method call works for either type at runtime.

5.Object Model:

Class	Description
RestaurantGUI	Main driver with main() and Swing window; handles all user events.
Fooditem(Abstract)	Base class with itemName, itemPrice; declares abstract totalPrice().
Main Course	Extends Fooditem; implements totalPrice() returning the stored price.
Dessert	Extends Fooditem; implements totalPrice() returning stored price.
Order	Stores orderId and total; addItem(double price) accumulates bill.
Restaurant	Holds name, location, totalSales; addOrder(double amount) updates sales.
Chef	Stores name and specialty; cook() prints cooking message.

Test class	AbstractionTest, Nila, Marzia, Drive – separate test files, not part of main GUI.

6.GUI Screenshot



7.Conclusion:

The Restaurant Management System delivers a working food-ordering GUI using Java and Swing. It fully applies encapsulation, inheritance, polymorphism, and abstraction, separates GUI from logic, supports real-time order building, simulates staff roles, and tracks sales within a session.